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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



✓ B. Tech ~~Summer~~ / Winter - 2025
(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering

Sem:- V

Subject: Internet of Things

Subject Code: CS5T003

Marks: 100

Date of Exam: 03/11/2025 Time: 3 Hrs.

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

CO 1: Identify the different technology.

CO 2: Apply IoT to different application

CO 3: Analysis and evaluate protocol used in IoT.

CO 4: Design and develop smart city in IoT.

CO 5: Analysis and evaluate the data received through sensors in IoT.

SECTION A

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Internet of Things (IoT)	L1/CO 1	2
Q.2	List out layers of the IoT three-layer architecture.	L1/CO 2	2
Q.3	Give example of cloud computing in IoT.	L2/CO 3	2
Q.4	Classify IoT LAN based on network protocol.	L2/CO 2	2
Q.5	Identify and define a sensor used for measuring distance.	L2/CO 1	2
Q.6	Define any two short-range communication protocols used in IoT.	L1/CO 3	2
Q.7	State the purpose of fog computing in IoT.	L1/CO 2	2
Q.8	Write the use of actuators in IoT systems.	L3/CO 5	2
Q.9	Identify the common challenge in IoT security.	L1/CO 4	2
Q.10	State IoT Data management and compute stack in IoT	L1/CO 1	2
Q.11	State the data analytics is pivotal for several reasons.	L2/CO 4	2
Q.12	Identify the function of the Arduino IDE.	L1/CO 5	2

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Categorize & explain the Different Applications of IoT.	L4/CO 1	4
Q.14	Discuss the Big Data Analytics with example.	L3/CO 4	4
Q.15	Illustrate the service models of cloud computing used in IoT.	L3/CO 2	4
Q.16	Examine MQTT protocol Architecture.	L4/CO 3	4
Q.17	Write short note on components of IoT Data Analytics	L3/CO 4	4
Q.18	Identify four common IoT applications built with Arduino.	L3/CO 5	4

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate Exclusive pair communication model in detail.	L4/CO 1	7
Q.20	Explain the five-layer architecture of IoT with a diagram.	L5/CO 2	7
Q.21	Describe the hardware layout of Raspberry Pi using a labelled diagram.	L5/CO 5	7
Q.22	Examine Rest based communication API used in IoT	L4/CO 1	7
Q.23	Describe the Communication API in IoT System	L6/CO 2	7
Q.24	Discuss the Big Data Analytics technologies in the IoT ecosystem.	L5/CO 4	7

SECTION D

1. Solve any TWO Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Design a simple IoT system using MQTT to monitor and control temperature in a building. How would MQTT facilitate communication between components?	L5/CO 3	12
Q.26	Explain the logical design of IoT in details.	L5/CO 1	12
Q.27	Describe the hardware layout of Raspberry Pi with a labelled diagram and explain the function of each component.	L6/CO 5	12